Algebraic Cycles by Fumiaki Suzuki

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This is a class given by Professor Suzuki in the spring semester of 2026 at Peking University. Here is the link to the course webpage: https://fumiaki-suzuki.github.io/spring2026. I copy the notes from the blackboard and write them in LaTeX. The notes are not complete, and I will update them when I have time. If you have any questions or find any mistakes, please feel free to contact me at 10211510098@stu.ecnu.edu.cn.

I use the following notations and conventions in the notes:

- $\mathbf{k}$ to denote the base field, and $\Bbbk$ is its algebraic closure;
- let X be a variety over $\mathbf{k}$, and $x \in X$ is a point, we denote by $\mathscr{K}(X)$ the function field of X, and $\kappa(x)$ the residue field of x;
- every scheme is assumed to be separated and of finite type over $\mathbf{k}$ unless otherwise specified;
- every variety is assumed to be integral;
- a subvariety of X is assumed to be closed unless otherwise specified.

1 March 03, 2026: Rational equivalence and Chow groups

Plan

I Basis of algebraic cycles: algebraic cycles, rational equivalence, Chow groups, chern and Segre classes, Gysin maps, intersection products, Grothendieck-Riemann-Roch, Motive.(Fulton)

II Higher Chow groups and motivic cohomology (Bloch, Mazza-Voevodsky-Weibel)

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1.1 Rational equivalence

A scheme will mean a separate scheme of finite type over a field $\mathbf{k}$ unless otherwise specified. A variety will mean an integral scheme.

Example 1.1. The closed subscheme X of $\mathbb{A}^n$ are in one-to-one correspondence with ideals I of $\mathbf{k}[x_1, \dots, x_n]$.

Example 1.2. A base change of a variety X may not be a variety. For example, let $X = \operatorname{Spec} \mathbb{R}[X]/(X^2+1)$ be a variety over $\mathbb{R}$. Then the base change $X_{\mathbb{C}} = X \times_{\operatorname{Spec} \mathbb{R}} \operatorname{Spec} \mathbb{C}$ is not a variety since it is not irreducible.

Definition 1.3. An i-cycle on a scheme X is a formal finite sum $\sum n_j[V_j]$ where $n_j \in \mathbb{Z}$ and V_j are closed subvarieties of X of dimension i.

Definition 1.4.

$$Z_i(X) \coloneqq \{\text{all } i\text{-cycles on } X\}$$

is the free abelian group generated by the i-dimensional closed subvarieties of X.

Definition 1.5. Let X be a variety of dimension n and $f \in \mathscr{K}(X)^\times$ be a nonzero rational function on X. Then define the divisor of f to be

$$\operatorname{div}(f) \coloneqq \sum_V \operatorname{ord}_V(f)[V] \in Z_{n-1}(X)$$

where the sum is taken over all codimension 1 subvarieties V of X and $\operatorname{ord}_V(f)$ is the order of vanishing of f along V after normalizing X. Explicitly, if $\nu : \tilde{X} \to X$ is the normalization of X, then

$$\operatorname{ord}_V(f) \coloneqq \sum_{\tilde{V} \subset \tilde{X}, \nu(\tilde{V}) = V} [\mathscr{K}(\tilde{V}) : \mathscr{K}(V)] \operatorname{ord}_{\tilde{V}}(f)$$

where the sum is taken over all codimension 1 subvarieties $\tilde{V}$ of $\tilde{X}$ that map to V under ν.

Example 1.6. Let $f \in \mathbf{k}[t]$ be a nonzero polynomial of degree d. By prime factorization, write $f = \prod_{i=1}^r f_j^{m_j}$ where f_j are distinct irreducible polynomials and m_j are positive integers. Regard f as a rational function on $\mathbb{P}^1$。 Then the divisor of f is

$$\operatorname{div}(f) = \sum_{j=1}^r m_j[V(f_j)] - d[\infty],$$

where $V(f_j)$ is the closed point of $\mathbb{P}^1$ defined by f_j and ∞ is the point at infinity of $\mathbb{P}^1$.

Definition 1.7. Let X be a scheme. A i-cycles $\alpha \in Z_i(X)$ are *rationally equivalent to zero* if there exists finitely many subvarieties W_j of X of dimension $i+1$ and rational functions $f_j \in \mathscr{K}(W_j)^\times$ such that

$$\alpha = \sum_j \operatorname{div}(f_j).$$

For example, consider the 0-cycle on X. There are finitely many closed points P_j of X and integers

n_j such that $\alpha = \sum n_j[P_j]$. Then α is rationally equivalent to zero if and only if there are finitely many curves C_k on X and rational functions $f_k \in \mathscr{K}(C_k)^\times$ such that

$$\alpha = \sum_k \operatorname{div}(f_k).$$

Definition 1.8. The *Chow group* of i-cycles on X is defined to be the quotient group

$$\mathrm{CH}_i(X) \coloneqq Z_i(X)/\{\text{rationally equivalent to zero}\}.$$

When X is smooth, we also write $\mathrm{CH}^i(X) \coloneqq \mathrm{CH}_{n-i}(X)$ where $n = \dim X$.

Example 1.9.

$$\mathrm{CH}_{n-1}(X) \cong \mathrm{Cl}(X).$$

$$\mathrm{CH}_*(X) \cong \mathrm{CH}_*(X_{\mathrm{red}}).$$

We will study the covariance and contravariance of Chow groups.

Proper pushforwards Let $f : X \to Y$ be a proper morphism of schemes, $V \subset X$ a subvariety of dimension i, and $W = f(V)$ the image of V in Y. Define

$$\deg(V/W) \coloneqq \begin{cases} [\mathscr{K}(V) : \mathscr{K}(W)], & \text{if } \dim V = \dim W, \\ 0, & \text{otherwise.} \end{cases}$$

For example, when $\mathbf{k} = \Bbbk$ of characteristic zero, if $\dim W = \dim V$, then $\deg(V/W)$ is the number of points in the fiber of $f|_V : V \to W$ over a general point of W. Define the pushforward of $[V]$ to be

$$f_*[V] \coloneqq \deg(V/W)[W].$$

Extend this definition linearly to get a homomorphism $f_* : Z_i(X) \to Z_i(Y)$. If $g : Y \to Z$ is another proper morphism, then $g_* \circ f_* = (g \circ f)_*$ because $\deg(V/(g \circ f)(V)) = \deg(V/W)\deg(W/g(W))$.

Example 1.10. Let X be a scheme and $f : X \to \operatorname{Spec} \mathbf{k}$ be the structure morphism. Define the degree map

$$\deg : Z_0(X) \to Z_0(\operatorname{Spec} \mathbf{k}) \cong \mathbb{Z}, \quad \sum n_j[P_j] \mapsto \sum n_j[\kappa(P_j) : \mathbf{k}].$$

Theorem 1.11. Let $f : X \to Y$ be a proper morphism and $\alpha \in Z_i(X)$. If α is rationally equivalent to zero on X, then $f_*\alpha$ is rationally equivalent to zero on Y. In particular, f_* induces a homomorphism $f_* : \mathrm{CH}_i(X) \to \mathrm{CH}_i(Y)$, i.e., CH_* is covariant w.r.t. proper morphisms.

Proof. May assume $\alpha = \operatorname{div}(r)$ for some rational function r on a subvariety V of X of dimension $i+1$. WTS: $f_*\alpha \sim 0$. Replace X by V and Y by $W \coloneqq f(V)$. Then $f : V \to W$ is a proper surjective morphism of varieties. The theorem follows from Proposition 1.12. □

Proposition 1.12. Let $f : X \to Y$ be a surjective proper morphism of varieties and $r \in \mathscr{K}(X)^\times$ a rational function. Then

(a) $f_* \operatorname{div}(r) = 0$ if $\dim W < \dim V$;

(b) $f_* \operatorname{div}(r) = \operatorname{div}(\mathrm{Nm}_{X/Y}(r))$ if $\dim W = \dim V$.

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Proof of (b). Let

$$S := \{y \in W : \dim f^{-1}(y) > 0\} \subset Y.$$

Then $\operatorname{codim}_Y S \geq 2$ since if $\operatorname{codim}_Y S = 1$, then $f^{-1}(S) = X$ which contradicts the surjectivity of f. Replace Y by $Y \setminus S$ and X by $f^{-1}(Y \setminus S)$. We may assume $f : X \to Y$ is a finite morphism of varieties.

Let $g : \tilde{X} \to X$ and $h : \tilde{Y} \to Y$ be the normalizations of X and Y respectively. Then we have the induced morphism $\tilde{f} : \tilde{X} \to \tilde{Y}$ such that $h \circ \tilde{f} = f \circ g$. Then $f, g, h, \tilde{f}$ are all finite morphisms of varieties and g, h are birational morphisms. We show that

(b1) for any rational function r on X, we have $g_* \operatorname{div}(r) = \operatorname{div}(r)$;

(b2) $\tilde{f}_* \operatorname{div}(r) = \operatorname{div}(\operatorname{Nm}_{X/Y}(r))$.

If (b1) and (b2) hold, then

$$f_* \operatorname{div}(r) = f_* g_* \operatorname{div}(r) = h_* \tilde{f}_* \operatorname{div}(r) = h_* \operatorname{div}(\operatorname{Nm}_{X/Y}(r)) = \operatorname{div}(\operatorname{Nm}_{X/Y}(r)).$$

Thus we are done.

For (b1), by definition, for a codimension 1 subvariety V of X, we have

$$\operatorname{ord}_V(r) = \sum_{\tilde{V} \subset \tilde{X}, g(\tilde{V}) = V} [\mathscr{K}(\tilde{V}) : \mathscr{K}(V)] \operatorname{ord}_{\tilde{V}}(r).$$

Taking the sum over all codimension 1 subvarieties V of X, we get

$$\operatorname{div}(r) = g_* \operatorname{div}(r).$$

For (b2), let V be a codimension 1 subvariety of $\tilde{Y}$. Then we need to show that

$$\operatorname{ord}_V(\operatorname{Nm}_{X/Y}(r)) = \sum_{\tilde{V} \subset \tilde{X}, \tilde{f}(\tilde{V}) = V} [\mathscr{K}(\tilde{V}) : \mathscr{K}(V)] \operatorname{ord}_{\tilde{V}}(r). \tag{1}$$

Set $A = \mathcal{O}_{\tilde{Y},V}$ and B be the integral closure of A in $\mathscr{K}(X)$. Then A is a DVR and B is f.g. torsion-free A-module. Hence $B \cong A^d$ for $d = [\mathscr{K}(X) : \mathscr{K}(Y)]$. Note that the maximal ideals of B are in one-to-one correspondence with the codimension 1 subvarieties $\tilde{V}$ of $\tilde{X}$ that map to V under $\tilde{f}$. May assume $r \in B$, then $B/rB \cong \prod_{\mathfrak{q} \subset B} B_{\mathfrak{q}}/rB_{\mathfrak{q}}$ is an artinian ring, where the product is taken over all maximal ideals $\mathfrak{q}$ of B. The RHS of (1) is equal to $\dim_{\mathscr{K}(V)} B/rB$. By the theory of elementary divisors,

$$\begin{tikzcd} B \arrow[r, "\text{times by } r"] \arrow[d, "\cong"] & B \arrow[d, "\cong"] \\ A^d \arrow[r, "M"] & A^d, \end{tikzcd}$$

where M is a diagonal matrix. Then

$$\dim_{\mathscr{K}(V)} B/rB = \dim_{\mathscr{K}(V)} A^d/(\det M)A^d = \text{ LHS of (1)}.$$

We are done. □

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Proof of (a). If $\dim Y < \dim X - 1$, the assertion is obvious. Assume $\dim Y = \dim X - 1$. Replacing Y by $\operatorname{Spec} \mathscr{K}(Y) = \operatorname{Spec} \mathbf{K}$ and X by $X \times_Y \operatorname{Spec} \mathscr{K}(Y)$, we may assume $Y = \operatorname{Spec} \mathscr{K}(Y)$ is a point and X is a proper curve. WTS: $f_* \operatorname{div}(r) = 0$, i.e., # of zeros of r equals # of poles of r counting multiplicity.

Let $g : \tilde{X} \to X$ be the normalization of X and $\tilde{f} : \tilde{X} \to Y$ be the induced morphism. Since $g_* \operatorname{div}(r) = \operatorname{div}(r)$, we only need to show that $\tilde{f}_* \operatorname{div}(r) = 0$. Choose any non-constant rational function t on $\tilde{X}$. It gives a finite morphism $t : \tilde{X} \to \mathbb{P}^1$. By (b), $\tilde{f}_* \operatorname{div}(r) = p_* \operatorname{div}(\operatorname{Nm}_{X/Y}(r))$, where $p : \mathbb{P}^1 \to \operatorname{Spec} \mathbf{K}$ is the structure morphism.

Write $\operatorname{div}(\operatorname{Nm}_{X/Y}(r)) = a/b$ with a, b coprime polynomials in $\mathbf{K}[t]$. By prime factorization, $a = \prod a_j^{m_j}$, $\div(a) = \sum m_j [V(a_j)] - (\sum m_j [\mathbf{K}(V(a_j)) : \mathbf{K}])[\infty]$. This means

$$p_* \operatorname{div}(a) = \sum m_j [\mathbf{K}(V(a_j)) : \mathbf{K}] - (\sum m_j [\mathbf{K}(\infty) : \mathbf{K}])[\mathbf{K}(\infty) : \mathbf{K}] = 0.$$

Similarly, $p_* \operatorname{div}(b) = 0$. Hence $p_* \operatorname{div}(\operatorname{Nm}_{X/Y}(r)) = p_* \operatorname{div}(a) - p_* \operatorname{div}(b) = 0$. □

Definition 1.13. Let X be a proper scheme over $\mathbf{k}$, $f : X \to \operatorname{Spec} \mathbf{k}$ be the structure morphism. Define the degree map

$$\deg : \mathrm{CH}_0(X) \to \mathrm{CH}_0(\operatorname{Spec} \mathbf{k}) \cong \mathbb{Z}, \quad \sum n_j [P_j] \mapsto \sum n_j [\kappa(P_j) : \mathbf{k}].$$

Example 1.14. Properness is necessary for the degree map of CH_0 to be well-defined. Let $X = \mathbb{A}^1$, $t \in \mathscr{K}(\mathbb{A}^1) = \mathbf{k}(t)$ be the coordinate function. Let $P = V(t) \subset \mathbb{A}^1$ be the divisor of t. On the one hand, $\deg([P]) = [\kappa(P) : \mathbf{k}] = 1$. On the other hand, $P = \operatorname{div}(t)$ is rationally equivalent to zero, so $\deg([P]) = 0$ if the degree map is well-defined.

2 March 05:

2.1 Length function

Let A be a ring, and M a f.g. A-module. By the Jordan-Hölder theorem, if M has a filtration

$$M = M_0 \supset M_1 \supset \cdots \supset M_r = 0$$

such that each $M_i / M_{i+1} \cong A/\mathfrak{m}_i$ for some maximal ideal $\mathfrak{m}_i$, then the length r does not depend on the choice of the filtration. We call the r the *length* of M, denoted by $\ell_A(M)$. Otherwise, we say $\ell_A(M) = \infty$.

If A an Artinian ring, then any f.g. A-module has finite length.

Example 2.1. Let A be a DVR, $r \neq 0 \in A$. Then $A/(r) \cong A/\mathfrak{m}^{\nu(r)}$, where ν is the valuation of A. Then we have a filtration

$$A/(r) = A/\mathfrak{m}^{\nu(r)} \supset A/\mathfrak{m}^{\nu(r)-1} \supset \cdots \supset A/\mathfrak{m} \supset 0$$

of length $\nu(r)$ with successive quotients isomorphic to $A/\mathfrak{m}$. So $\ell_A(A/(r)) = \nu(r)$.

Lemma 2.2 (See [Ful13, A.3.1])**.** Let A be a 1-dimensional local noetherian domain with $\mathbf{K} = \operatorname{Frac}(A)$. Let B be the integral closure of A in $\mathbf{K}$. Then B is a Dedekind domain. Assume B is a f.g.

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A-module, then $\forall r \neq 0 \in A$, we have

$$\ell_A(A/rA) = \sum_{\mathfrak{m} \in \operatorname{mSpec} B} \ell_{B_\mathfrak{m}}(B_\mathfrak{m}/rB_\mathfrak{m})[B/\mathfrak{m} : A/\mathfrak{m}_A].$$

Apply Lemma 2.2 to $A = \mathcal{O}_{X,V}$, where X is a variety and V is a subvariety of codimension 1. Then for all $r = a/b \in \mathscr{K}(X)^\times$ with $a, b \in \mathcal{O}_{X,V}$, we have

$$\operatorname{ord}_V(r) = \ell_{\mathcal{O}_{X,V}}(\mathcal{O}_{X,V}/(a)) - \ell_{\mathcal{O}_{X,V}}(\mathcal{O}_{X,V}/(b)).$$

Example 2.3. Let $X = \{y^2 = x^3\} \subset \mathbb{A}^2$ be a cuspidal cubic curve, and $O = \{(0,0)\}$ be the cusp. Then $\mathscr{K}(X) = \operatorname{Frac} \mathbf{k}[x,y]/(y^2 - x^3) \cong \operatorname{Frac} \mathbf{k}[t]$ via $x \mapsto t^2, y \mapsto t^3$. Compute:

$$\operatorname{ord}_O(x) = \ell_{\mathcal{O}_{X,O}}(\mathcal{O}_{X,O}/(x)) = \ell_{\mathcal{O}_{X,O}}(\mathbf{k}[x,y]/(x, y^2 - x^3)) = \dim_\mathbf{k} \mathbf{k}[x,y]/(x, y^2) = 2.$$

Hence

$$\operatorname{div}(x) = 2[O].$$

Similarly, we have

$$\operatorname{div}(y) = 3[O].$$

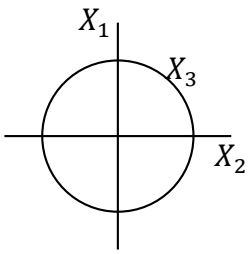
Definition 2.4. Let X be a scheme, $X_1, \dots, X_t$ be the irreducible components of X. Define the *fundamental cycle* of X to be

$$[X] = \sum_{j=1}^{t} \ell(\mathcal{O}_{X,X_j})[X_j].$$

The length $\ell(\mathcal{O}_{X,X_j})$ is called the *generic multiplicity* of X_j in X.

Here $\mathcal{O}_{X,X_j}$ is an Artinian local ring, so $\ell(\mathcal{O}_{X,X_j}) < \infty$.

Example 2.5. Let $X = \{x^3y^2(x^2 + y^2 - 1) = 0\} \subset \mathbb{A}^2$. Then X has three irreducible components $X_1 = \{x = 0\}, X_2 = \{y = 0\}, X_3 = \{x^2 + y^2 - 1 = 0\}$.



We have:

$$\ell(\mathcal{O}_{X,X_1}) = \ell\left(\left(\mathbf{k}[x,y]/(x^3y^2(x^2 + y^2 - 1))\right)_{(x)}\right) = \dim_{\mathbf{k}(y)} \mathbf{k}(y)[x]/(x^3) = 3.$$

Similarly, we have $\ell(\mathcal{O}_{X,X_2}) = 2$ and $\ell(\mathcal{O}_{X,X_3}) = 1$. Hence

$$[X] = 3[X_1] + 2[X_2] + [X_3].$$

Proposition 2.6. Let X be a scheme, $\alpha \in Z_i(X)$. Then $\alpha \sim 0$ if and only if there exist finitely many subvarieties of dimension $i + 1$ of $\mathbb{P}^1 \times X$, say $W_1, \dots, W_r$, such that the projection $W_j \to \mathbb{P}^1$ is

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dominant for all j, and

$$\alpha = \sum_{j=1}^{r} [f_j^{-1}(0)] - [f_j^{-1}(\infty)].$$

Ideal of Proof. For a variety V and a dominant morphism $f : V \to \mathbb{P}^1$, the divisor of f is just $\operatorname{div}(f) = [f^{-1}(0)] - [f^{-1}(\infty)]$.

For a subvariety $V' \subset X$ and $f' \in \mathscr{K}(V')^\times$, apply the above result to the closure of the graph Γ of f' in $V' \times \mathbb{P}^1$. Note that $\Gamma \to V'$ is birational. □

Example 2.7. As an application of Proposition 2.6, one can show

$$\mathrm{CH}_i(\mathbb{A}^n) = \begin{cases} \mathbb{Z} \cdot [\mathbb{A}^n], & i = n, \\ 0, & i < n. \end{cases}$$

(left as an exercise)

2.2 Flat pullbacks

Let $f : X \to Y$ be a flat morphism of schemes of relative dimension n. In this note, by a *flat morphism*, we always mean a flat morphism of constants relative dimension. Here are examples that we should keep in mind:

- an open embedding ($n = 0$);
- the projection of a vector bundle or a projective bundle;
- a dominant morphism from a variety to a smooth curve;
- a pure-dimensional scheme $X \to \operatorname{Spec} \mathbf{k}$.

For such f and a subvariety $V \subset Y$ of dimension i, we define

$$f^*[V] = [f^{-1}(V)] \in Z_{i+n}(X).$$

This gives us

$$f^* : Z_i(Y) \to Z_{i+n}(X).$$

Lemma 2.8. Let $f : X \to Y$ be a flat morphism of relative dimension n, $Z \subset Y$ be a subscheme. Then

$$f^*[Z] = [f^{-1}(Z)].$$

Proof. Let $V \subset Y$ be an irreducible component of Z, W be an irreducible component of $f^{-1}(V)$. The coefficient of $[V]$ in $[Z]$ (resp. $[W]$ in $[f^{-1}(V)]$) is $\ell(\mathcal{O}_{Z,V})$ (resp. $\ell(\mathcal{O}_{f^{-1}(V),W})$). Hence the coefficient of $[W]$ in $f^*[Z]$ is $\ell(\mathcal{O}_{Z,V}) \cdot \ell(\mathcal{O}_{f^{-1}(V),W})$. Note that

$$\mathcal{O}_{f^{-1}(V),W} = \mathcal{O}_{f^{-1}(Z),W}/(\mathfrak{m}_{Z,V}\mathcal{O}_{f^{-1}(Z),W}).$$

On the other hand, the coefficient of $[W]$ is $[f^{-1}(Z)]$ is $\ell(\mathcal{O}_{f^{-1}(Z),W})$. Hence the proof reduced to show

$$\ell(\mathcal{O}_{Z,V}) \cdot \ell(\mathcal{O}_{f^{-1}(Z),W}/(\mathfrak{m}_{Z,V}\mathcal{O}_{f^{-1}(Z),W})) = \ell(\mathcal{O}_{f^{-1}(Z),W}).$$

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Here $\mathcal{O}_{Z,V} \to \mathcal{O}_{f^{-1}(Z),W}$ is a flat local homomorphism of Artinian local rings, so the above formula follows from [Ful13, A.4.1]. □

By the above lemma, if $f : X \to Y$, $g : Y \to Z$ are flat morphisms of constants relative dimension, then $(g \circ f)^* = f^* \circ g^*$. Indeed, for a subvariety $V \subset Z$, we have

$$(g \circ f)^*[V] = [(g \circ f)^{-1}(V)] = [f^{-1}(g^{-1}(V))] = f^*[g^{-1}(V)] = f^* \circ g^*[V].$$

Here on the third equality, we use the above lemma.

Proposition 2.9. Let

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "g"] & Y \end{tikzcd}$$

be a Cartesian square of schemes, where f is proper and g is flat of constants relative dimension. Then for every $\alpha \in Z_\bullet(X)$, we have

$$g^* f_* \alpha = f'_* g'^* \alpha \in Z_\bullet(Y').$$

Proof.

Case 1. When f is a closed embedding.

For a subvariety $V \subset X$, we have

$$g^* f_*[V] = g^*[V] = [g^{-1}(V)] = [g'^{-1}(V)] = f'_*[g'^{-1}(V)] = f'_* g'^*[V].$$

Case 2. X, Y varieties, f is surjective and $\alpha = [X]$.

Then $f_*[X] = d[Y]$ for some $d \in \mathbb{Z}_{\geq 0}$. WTS $f'_*[X'] = d[Y']$. If $\dim Y < \dim X$, then $\dim Y' < \dim X'$ and $d = 0$, so the formula holds. Assume $\dim Y = \dim X$. Then $d = \deg(X/Y)$ is the degree of the field extension $\mathscr{K}(X)/\mathscr{K}(Y)$. The formula may be checked on the generic point of Y. Hence we may assume $Y = \operatorname{Spec} \mathbf{K}$ with $\mathbf{K} = \mathscr{K}(Y)$ and $X = \operatorname{Spec} \mathbf{L}$ with $\mathbf{L} = \mathscr{K}(X)$. Here $\mathbf{L}/\mathbf{K}$ is a finite extension of fields of degree d. The diagram becomes

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"'] & \operatorname{Spec} \mathbf{L} \arrow[d, "f"] \\ Y' \arrow[r, "g"] & \operatorname{Spec} \mathbf{K}. \end{tikzcd}$$

Let V be an irreducible component of Y'. The coefficient of $[V]$ in $f'_*[X']$ is

$$\sum_W \ell(\mathcal{O}_{X',W}) \cdot \deg(W/V),$$

where W runs through the irreducible components of X' that dominate V. The formula follows from:

$$[\mathbf{L} : \mathbf{K}]\ell(\mathcal{O}_{Y',V}) = \sum \ell(\mathcal{O}_{X',W})[\mathscr{K}(W) : \mathscr{K}(V)],$$

(see [Ful13, A.1.3].)

Case 3. The general case.

It follows from the above two cases by diagram chasing.

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Theorem 2.10. Let $f : X \to Y$ be a flat morphism of relative dimension n and $\alpha \in Z_i(Y)$. If $\alpha \sim 0$ on Y, then $f^*\alpha \sim 0$ on X. In particular, we have a well-defined pullback homomorphism

$$f^* : \mathrm{CH}_i(Y) \to \mathrm{CH}_{i+n}(X).$$

i.e., CH_* is contravariant functorial with respect to flat morphisms.

Proof. By Proposition 2.6, may assume $\alpha = [g^{-1}(0)] - [g^{-1}(\infty)]$ for V an $i+1$-dimensional subvariety of $\mathbb{P}^1 \times Y$ such that the projection $g : V \to \mathbb{P}^1$ is dominant. In particular, g is flat. We have a diagram:

$$\begin{tikzcd}
W \arrow[r, hook] \arrow[d] \arrow[ddr, bend right, "h"'] & \mathbb{P}^1 \times X \arrow[r, "q"] \arrow[d, "\mathrm{id}\times f"] & X \arrow[d, "f"] \\
V \arrow[r, hook] \arrow[dr, "g"'] & \mathbb{P}^1 \times Y \arrow[r, "p"] \arrow[d] & Y \\
& \mathbb{P}^1 &
\end{tikzcd}$$

Then

$$\begin{aligned}
f^*\alpha &= f^*p_*([g^{-1}(0)] - [g^{-1}(\infty)]) \\
&= q_*(\mathrm{id} \times f)^*([g^{-1}(0)] - [g^{-1}(\infty)]) \\
&= q_*([h^{-1}(0)] - [h^{-1}(\infty)]).
\end{aligned}$$

Let $W_1, \dots, W_t$ be the irreducible components of W, $h_j = h|_{W_j}$. Suppose that $[W] = \sum_{j=1}^t m_j[W_j]$. By Lemma 2.11, we have $[h^{-1}(0)] = \sum_{j=1}^t m_j[h_j^{-1}(0)]$ and $[h^{-1}(\infty)] = \sum_{j=1}^t m_j[h_j^{-1}(\infty)]$. Hence

$$f^*\alpha = m_j q_*\left(\sum_{j=1}^t [h_j^{-1}(0)] - [h_j^{-1}(\infty)]\right) = \sum_{j=1}^t m_j q_*([h_j^{-1}(0)] - [h_j^{-1}(\infty)]) \sim 0.$$

We are done. □

Lemma 2.11. Let X be a purely n-dimensional scheme, $X_1, \dots, X_t$ be the irreducible components of X. Write $[X] = \sum_{j=1}^t m_j[X_j]$ with $m_j = \ell(\mathcal{O}_{X,X_j})$. Let D be a Cartier divisor on X. Then, let $D_j = D|_{X_j}$, we have

$$[D] = \sum_{j=1}^t m_j[D_j].$$

Proof. Let V be a codimension 1 subvariety of X, $A = \mathcal{O}_{X,V}$, $r \in A$ be a local equation of D. The coefficient of $[V]$ in $[D]$ is $\ell(\mathcal{O}_{D,V}) = \ell(A/(r))$. On the other hand, the coefficient of $[V]$ in $[D_j]$ is $\ell(\mathcal{O}_{D_j,V}) = \ell(A/(P_j + rA))$, where P_j is the prime ideal of A corresponding to X_j. The formula follows from:

$$\ell(A/(r)) = \sum_{j=1}^t \ell(A_{P_j}) \cdot \ell(A/(P_j + rA)).$$

(See [Ful13, A.2.7].) □

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2.3 Localization exact sequence

Proposition 2.12. Let X be a scheme, $Y \subset X$ be a closed subscheme, $U = X \setminus Y$. Denote by $i_Y : Y \hookrightarrow X$ and $j_U : U \hookrightarrow X$ the embeddings. Then we have an exact sequence

$$\mathrm{CH}_i(Y) \xrightarrow{i_{Y*}} \mathrm{CH}_i(X) \xrightarrow{j_U^*} \mathrm{CH}_i(U) \to 0.$$

Remark 2.13. The surjectivity on the right is almost never true in the topological setting as a topological cycle on U does NOT lift to X.

Ideal of Proof. We have

$$Z_i(Y) \xrightarrow{i_{Y*}} Z_i(X) \xrightarrow{j_U^*} Z_i(U) \to 0$$

is exact. (For $V \subset U$ subvariety of dimension i, its closure $\overline{V}$ in X is a subvariety of dimension i and $j_U^*[\overline{V}] = [V]$.)

Passing to the quotient, we get a complex

$$\mathrm{CH}_i(Y) \xrightarrow{i_{Y*}} \mathrm{CH}_i(X) \xrightarrow{j_U^*} \mathrm{CH}_i(U) \to 0.$$

We only need to show that if $\alpha \in Z_i(X)$ and $j_U^*\alpha \sim 0$ on U, then $\alpha \in i_{Y*}Z_i(Y) + \sim$ on X. We can assume that $j_U^*\alpha = \mathrm{div}(f)$ for some $f \in \mathscr{K}(V)^\times$ with $V \subset U$ a subvariety of dimension $i+1$. Consider f as a rational function on $\overline{V}$. Then $\alpha - \mathrm{div}(f) = i_{Y*}\beta \in Z_i(X)$ for some $\beta \in Z_i(Y)$ by the exact sequence on the level of cycles. Hence $\alpha \sim i_{Y*}\beta$ on X and we are done. □

Example 2.14. Let $L_i \subset \mathbb{P}^n$ be a linear subspace of dimension i for $i = 0, \dots, n$. Then $\mathrm{CH}_i(\mathbb{P}^n)$ is generated by $[L_i]$ for $i = 0, \dots, n$. Indeed, we have an exact sequence

$$\mathrm{CH}_i(L_{n-1}) \xrightarrow{\iota_*} \mathrm{CH}_i(\mathbb{P}^n) \xrightarrow{j^*} \mathrm{CH}_i(\mathbb{A}^n) \to 0.$$

The assertion follows from $\mathrm{CH}_i(\mathbb{A}^n) = 0$ for $i < n$ and $\mathrm{CH}_n(L_{n-1}) = 0$ for $i = n$. We will see later that $\mathrm{CH}_i(\mathbb{P}^n) = \mathbb{Z} \cdot [L_i]$.

3 March 10: divisors

A goal for us in the coming weeks is to define intersection products for Chow groups.

Example 3.1. Let X be a topological space. The cap product is defined on singular homology and cohomology on X as follows:

$$H_i(X, \mathbb{Z}) \otimes H^j(X, \mathbb{Z}) \to H_{i-j}(X, \mathbb{Z}), \quad (\sigma, \psi) \mapsto \psi(\sigma).$$

The cohomology group $H^j(X, \mathbb{Z})$ may be regarded as the collection of operations on $H_\bullet(X, \mathbb{Z})$ with grading shift $-j$.

For a scheme X, we will define

$$\mathrm{CH}_i(X) \otimes \text{“}\,\mathrm{CH}^j(X)\text{”} \to \mathrm{CH}_{i-j}(X),$$

where “$\mathrm{CH}^j(X)$” contains the j-th Chern class of vector bundles on X. The simplest case is $j = 1$,

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corresponding to line bundles or Cartier divisors.

3.1 Divisors

Definition 3.2. Let X be a n-dimensional variety. A *Weil divisor* on X will mean an $(n-1)$-cycle on X. A *Cartier divisor* on X is defined by data (U_α, f_α) where U_α is an open cover of X and f_α is a non-zero rational function on U_α such that f_α/f_β is a unit on $U_\alpha \cap U_\beta$. In other words, it is an element of

$$\mathrm{Div}(X) := \Gamma(X, \mathscr{K}(X)^*/\mathcal{O}_X^*).$$

A Cartier divisor is called *principal* if it lies in the image of the natural map

$$\Gamma(X, \mathscr{K}(X)^*) \to \mathrm{Div}(X).$$

Two Cartier divisors are called *linearly equivalent* if their difference is principal, and we write $D \sim D'$.

Note

$$0 \to \mathcal{O}_X^* \to \mathscr{K}(X)^* \to \mathrm{Div}(X) \to 0$$

is exact, hence we have the induced map

$$\mathrm{Div}(X) \to \mathrm{Pic}(X) \cong H^1(X, \mathcal{O}_X^*).$$

We have the natural map $\mathrm{Div}(X) \to Z_{n-1}(X)$ sending a Cartier divisor $\{(U_\alpha, f_\alpha)\}$ to a Weil divisor $\sum_V \mathrm{ord}_V(f_\alpha) \cdot V$, where V runs through all irreducible subvarieties of codimension 1 in X and $\mathrm{ord}_V(f_\alpha)$ is the order of vanishing of f_α along V whenever $V \cap U_\alpha \neq \varnothing$. Since f_α/f_β is a unit on $U_\alpha \cap U_\beta$, $\mathrm{ord}_V(f_\alpha) = \mathrm{ord}_V(f_\beta)$, so the above definition is independent of the choice of α.

Example 3.3. If X is a smooth variety, $\mathrm{Pic}(X) \cong \mathrm{CH}_{n-1}(X)$ (see [Har77, Chapter II, Section 6]). In general, the map $\mathrm{Pic}(X) \to \mathrm{CH}_{n-1}(X)$ is neither injective nor surjective. For instance, $X := \{xy = z^2\} \subset \mathbb{A}^3$ and $l := \{x = z = 0\} \subset X$ is a Weil divisor that is not Cartier. And $\mathrm{CH}_1(X) = \mathrm{Cl}(X) = \mathbb{Z}/2$ generated by $[l]$. This implies $\mathrm{Pic}(X) \to \mathrm{CH}_1(X)$ is the zero map. In fact, this is injective (using X is normal), so $\mathrm{Pic}(X) = 0$.

Note that the definition of Cartier divisors extends to schemes. (But $(\mathrm{Div}(X)/\sim) \to \mathrm{Pic}(X)$ is NOT an isomorphism in general!)

3.2 Intersection with divisors

Line bundles always can be pull-backed, so do Cartier divisors modulo linear equivalence.

Example 3.4. Let $\pi : \widetilde{\mathbb{A}^n} \to \mathbb{A}^n$ be the blow-up of $\mathbb{A}^n$ at the origin, and let $\mathbb{P}^{n-1} \cong E \subset \widetilde{\mathbb{A}^n}$ be the exceptional divisor. Then $\mathcal{O}_{\mathbb{A}^n}(E)|_E = \mathcal{N}_{E/\widetilde{\mathbb{A}^n}} = \mathcal{O}_E(-1)$, but E does NOT pull-back to E.

For a Cartier divisor D on a scheme X, define the *support* of D to be

$$|D| := \mathrm{Supp}(D) := \bigcup_{V \subset X, \mathrm{ord}_V(D) \neq 0} V.$$

Cartier divisors have supports , but line bundles do NOT.

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Definition 3.5. A *pseudo-divisor* on X is a triple (L, Z, s) where

- L is a line bundle on X,
- $Z \subset X$ is a closed subset of X,
- s is a nowhere vanishing section on $L|_{X-Z}$.

A pseudo-divisor is either

- $(L, X, 1)$ for some line bundle L on X, or
- $(\mathcal{O}_X(D), |D|, s_D)$ for some Cartier divisor D on X, where s_D corresponds to $1 \in \mathscr{K}(X)^*$ with $\mathcal{O}_X(D) \subset \mathscr{K}(X)^*$.

We also denote a pseudo-divisor D and the corresponding triple by $(L_D, |D|, s_D)$.

Definition 3.6. For pseudo-divisors D and D', define

$$D + D' \coloneqq (L_D \otimes L'_D, |D| \cup |D'|, s_D \otimes s'_D),$$
$$-D \coloneqq (L_D^{-1}, |D|, s_D^{-1}).$$

For a morphism $f : Y \to X$, define $f^*D \coloneqq (f^*L_D, f^{-1}(|D|), f^*s_D)$.

Definition 3.7. When X is an n-dimensional variety, define $[D] \in \mathrm{CH}_{n-1}(|D|)$ to be $c_1(L_D)|_{|D|}$.

Definition 3.8. Let X be a scheme, D be a pseudo-divisor on X, $V \subset X$ be a subvariety of X of dimension i with inclusion $j : V \to X$. We define

$$D \cdot [V] \coloneqq [j^*D] \in \mathrm{CH}_{i-1}(|D| \cap V).$$

For an i-cycle $\alpha = \sum n_V[V]$ on X, define

$$D \cdot \alpha \coloneqq \sum n_V(D \cdot [V]) \in \mathrm{CH}_{i-1}(|D| \cap |\alpha|),$$

where $|\alpha| \coloneqq \bigcup_{n_V \neq 0} V$ is the support of α.

Example 3.9. Let X be a purely n-dimensional scheme, D be a Cartier divisor on X. Then $D \cdot [X] = [D]$ by Lemma 2.11.

Proposition 3.10. We have the following basic properties.

(a) Let X be a scheme, D, D' be pseudo-divisors on X, α, α' be i-cycles on X. Then

$$D \cdot (\alpha + \alpha') = D \cdot \alpha + D \cdot \alpha' \in \mathrm{CH}_{i-1}(|D| \cap (|\alpha| \cup |\alpha'|)).$$

(b) The same as (a). We have

$$(D + D') \cdot \alpha = D \cdot \alpha + D' \cdot \alpha \in \mathrm{CH}_{i-1}\big((|D| \cup |D'|) \cap |\alpha|\big).$$

(c) Let $f : X \to Y$ be a proper morphism, α be an i-cycle on X, D be a pseudo-divisor on Y, then $f_*(f^*D \cdot \alpha) = D \cdot f_*\alpha \in \mathrm{CH}_{i-1}(|D| \cap f(|\alpha|))$.

(d) Let $f : X \to Y$ be a flat morphism of relative dimension n, α be an i-cycle on Y, D be a pseudo-divisor on Y, then

$$f^*(D \cdot \alpha) = f^*D \cdot f^*\alpha \in \mathrm{CH}_{i+n-1}(f^{-1}(|D|) \cap f^{-1}(|\alpha|)).$$

(e) Let D be a pseudo-divisor on a scheme X s.t. $L_D \cong \mathcal{O}_X$. Then, for all i-cycles α on X, $D \cdot \alpha = 0 \in \mathrm{CH}_{i-1}(|\alpha|)$.

Proof. (a) By definition.

Hence in (b)-(e), may assume $\alpha = [V]$ for some variety V.

(b) By linearity of restricting pseudo-divisors to a subvariety and taking the class of pseudo-divisors.

(c) If f is a closed embedding, it is ok. In general, may assume: X is a variety, $\alpha = [X]$, $f : X \to Y$ is surjective proper, D is a Cartier divisor. WTS, $f_*f^*[D] = \deg(X/Y)[D]$. By working locally, may assume $D = \mathrm{div}(r)$ for some $r \in \mathscr{K}(X)^\times$, then

$$f_*f^* \operatorname{div}(r) = \operatorname{div}(r^d) = d \operatorname{div}(r),$$

where $d = \deg(X/Y)$.

(d) May assume: Y is a variety and $\alpha = [Y]$, WTS: $f^*[D] = [f^{-1}(D)]$. It is by Lemma 2.8.

(e) May assume X is a variety, $\alpha = [X]$, $D = \mathrm{div}(r)$ for some $r \in \mathscr{K}(X)^\times$, then $\mathrm{div}(r) \cdot [X] = \mathrm{div}(r) = 0 \in \mathrm{CH}_{i-1}(X)$. □

Theorem 3.11. Let D, D' be Cartier divisors on an n-dimensional variety X. Then

$$D \cdot [D'] = D' \cdot [D] \in \mathrm{CH}_{n-2}(|D| \cap |D'|).$$

Proof.

Case 1. D, D' are effective Cartier divisors (i.e. codimension 1 closed subschemes, locally defined by single non zero functions).

By induction on:

$$\mathcal{E}(D, D') := \max\{\mathrm{ord}_V(D)\,\mathrm{ord}_V(D') \mid V \subset X \text{ of codimension } 1\}.$$

The basis case is $\mathcal{E}(D, D') = 0$, i.e. D and D' intersect properly. Let $W \subset X$ a codimension 2 subvariety, $A := \mathcal{O}_{X,W}$, $a, a' \in A$ be local equations of D and D'. Let $V \subset X$ be a codimension 1 subvariety containing W which correspond to a height 1 prime ideal $\mathfrak{p} \subset A$. Then the

- coefficient of $[V]$ in $[D']$ is $\ell(\mathcal{O}_{D',V}) = \ell_{A_\mathfrak{p}}(A_\mathfrak{p}/a'A_\mathfrak{p})$, and
- coefficient of $[W]$ in $D \cdot [V]$ is $\ell(\mathcal{O}_{D \cap V,W}) = \ell_{A/\mathfrak{p}}(A/(\mathfrak{p} + aA))$.

Hence the coefficient of $[W]$ in $D \cdot [D']$ is

$$\sum_\mathfrak{p} \ell_{A_\mathfrak{p}}(A_\mathfrak{p}/a'A_\mathfrak{p})\ell_{A/\mathfrak{p}}(A/(\mathfrak{p} + aA)).$$

By [Ful13, A 2.7 and A 2.8], change a and a' is ok. By symmetry, $D \cdot [D'] = [D'] \cdot [D]$ as cycles.

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Now we assume $\mathcal{E}(D, D') > 0$ (just an idea). Let $\pi : \tilde{X} \to X$ be the blow-up along $D \cap D'$, $E \subset \tilde{X}$ be the exceptional divisor. Then $\pi^* D = \tilde{D} + E, \pi^* D' = \tilde{D}' + E$, where $\tilde{D}$ and $\tilde{D}'$ are effective Cartier divisors such that $\tilde{D} \cap \tilde{D}' = \varnothing$. Moreover, $\mathcal{E}(\tilde{D}, E), \mathcal{E}(\tilde{D}', E) < \mathcal{E}(D, D')$ (for the proof, read Fulton). Then

$$D \cdot [D'] = \pi_*((\tilde{D} + E) \cdot ([\tilde{D}'] + [E])) = \pi_*((\tilde{D}' + E) \cdot ([\tilde{D}] + [E])) = D' \cdot [D].$$

Case 2. D is arbitrary, D' is effective.

If D is the difference of 2 effective Cartier divisors, the proof is reduced to case 1. However, this is in general FALSE.

In general, may assume: $X = \operatorname{Spec} A$, $a \in \mathscr{K}(X)^*$ be the local equation for D. Define $I \coloneqq \{b \in A \mid a \cdot b \in A\}$ be the ideal of denominations of a. Let $\pi : \tilde{X} \to X$ be the blow-up along $V(I)$, E be the exceptional divisor. Then $\pi^* D = C - E$, where C is an effective Cartier divisor. Since the Theorem holds for $(C, \pi^* D'), (E, \pi^* D')$, it also holds for (D, D').

Case 3. D, D' are arbitrary.

Let $\pi : \tilde{X} \to X$ as Case 2. Since the Theorem holds for $(C, \pi^* D')$ and $(E, \pi^* D')$. It also holds for (D, D'). □

Yang: I need more computation for the theorem, in particular the blow up trick.

Corollary 3.12. Let X be a scheme, α be an i-cycle on X. If $\alpha \sim 0$ on X, then for all pseudo-divisor D on X, $D \cdot \alpha = 0 \in \mathrm{CH}_{i-1}(|D| \cap |\alpha|)$. In particular, we have well defined

$$\mathrm{CH}_i(X) \to \mathrm{CH}_{i-1}(|D|), \quad \alpha \mapsto D \cdot \alpha.$$

Proof. May assume: X is a variety, $\alpha = \operatorname{div}(f)$ for some $f \in \mathscr{K}(X)^*$. Then $D \cdot \operatorname{div}(f) = \operatorname{div}(f) \cdot [D] = 0 \in \mathrm{CH}_{n-2}(|D| \cap |\alpha|)$. □

Definition 3.13. Let X be a scheme, L be a line bundle on X. For an i-dimensional subvariety V of X, define

$$c_1(L) \cap [V] \coloneqq \text{the class of } L|_V \in \mathrm{CH}_{i-1}(V).$$

By Corollary 3.12, we have well defined

$$c_1(L) \cap : \mathrm{CH}_i(X) \to \mathrm{CH}_{i-1}(X).$$

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